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Taller 2
Sistemas Dinámicos y Control
Respuesta en Frecuencia

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Problemas del libro “Feedback Control of Dynamic Systems – Eight Edition – Franklin, Powell, Emami-Naeini, Pearson”.

Capítulo 6: Respuesta en frecuencia

Control del error en estado estacionario

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Estabilidad neutral

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Compensación

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6.6 *Multiple poles at the origin*. Sketch the asymptotes of the Bode plot magnitude and phase for each of the following open-loop transfer functions. After completing the hand sketches, verify your results using Matlab. Turn in your hand sketches and the Matlab results on the same scales.

(a) $L(s) = \frac{1}{s^2(s+0.7)}$

(b) $L(s) = \frac{100}{s^3(s+80)}$

(c) $L(s) = \frac{1}{s^4(2s+5)}$

(d) $L(s) = \frac{s+7.5}{s^2(s+75)}$

(e) $L(s) = \frac{1.5s+1}{s^3(s+0.1)}$

(f) $L(s) = \frac{(s+9)^2}{s^3(s+50)}$

(g) $L(s) = \frac{(s+0.6)^2}{s^3(s+1.3)^2}$

6.7 *Mixed real and complex poles*. Sketch the asymptotes of the Bode plot magnitude and phase for each of the following open-loop transfer functions. After completing the hand sketches verify your results using Matlab. Turn in your hand sketches and the Matlab results on the same scales.

(a) $L(s) = \frac{(s+0.5)}{s(5s+1)(s^2+0.2s+0.6)}$

(b) $L(s) = \frac{(2s+4)}{s^2(s+8)(s^2+5s+27)}$

(c) $L(s) = \frac{(s+0.75)^2}{s^2(1.2s+8.1)(s^2+5s+27)}$

(d) $L(s) = \frac{(s+50)(2s^2+5s+4)}{s^2(s+5)(s^2+60s+120)}$

(e) $L(s) = \frac{[(s+2)^2+2]}{s^2(s^2+7s+5)}$

6.8 *Right half-plane poles and zeros*. Sketch the asymptotes of the Bode plot magnitude and phase for each of the following open-loop transfer functions. Make sure the phase asymptotes properly take the RHP singularity into account by sketching the complex plane to see how the $\angle L(s)$ changes as s goes from 0 to $+j\infty$. After completing the hand sketches verify your results using Matlab. Turn in your hand sketches and the Matlab results on the same scales.

(a) $L(s) = \frac{s+4}{s+12}\frac{1}{s^2-9}$; (The model for a case of magnetic levitation with lead compensation.)

(b) $L(s) = \frac{s+4}{s(s+6)}\frac{1}{s^2-22}$; (The magnetic levitation system with integral control and lead compensation.)

(c) $L(s) = \frac{11s-7}{s^2}$

(d) $L(s) = \frac{s^2+4s+3}{s(s+2.5)^2(s^2-3s+5)}$

(e) $L(s) = \dfrac{(s+7.5)}{s(s-0.5)(s+20)^2}$

(f) $L(s) = \dfrac{1}{(s-9)[(s+2)^2+5)]}$

6.9 A certain system is represented by the asymptotic Bode diagram shown in Fig. 6.85. Find and sketch the response of this system to a unit step input (assuming zero initial conditions).

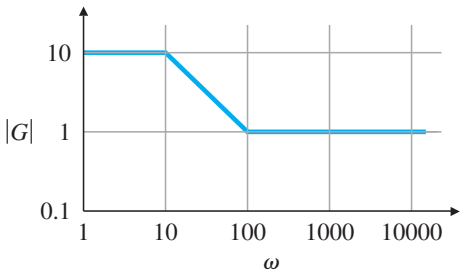


Figure 6.85
Magnitude portion of Bode plot for Problem 6.9.

6.10 Prove that magnitude slope of -2 in a Bode plot corresponds to -40 db per decade or -12 db per octave.

6.11 A second-order system with a damping ratio $\zeta = 0.6$ and an additional zero is given by

$$G(s) = \frac{\left(\frac{s}{\alpha}+1\right)}{s^2+1.2s+1}.$$

Use Matlab to compare the M_p from the step response of the system for $a = 0.01$, 0.1, 1, 10, and 100 with the M_r from the frequency response of each case. Is there a correlation between M_r and M_p?

6.12 A second order system with $\zeta = 0.6$ and an additional pole is given by

$$G(s) = \frac{2}{\left[\left(\frac{s}{p}\right)+1\right](s^2+1.2\sqrt{2}s+2)}$$

Draw Bode plots with $p = 0.01$, 0.1, 1, 10, and 100. What conclusions can you draw about the effect of an extra pole on the bandwidth compared to the bandwidth for the second-order system with no extra pole?

6.13 For the closed-loop transfer function

$$T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2},$$

derive the following expression for the bandwidth ω_{BW} of $T(s)$ in terms of ω_n and ζ:

$$\omega_{\text{BW}} = \omega_n\sqrt{1-2\zeta^2+\sqrt{2+4\zeta^4-4\zeta^2}}.$$

Assuming $\omega_n = 1$, plot ω_{BW} for $0 \le \zeta \le 1$.

6.14 Consider the system whose transfer function is

$$G(s) = \frac{A_0\omega_0 s}{s^2+\frac{\omega_0}{Q}s+\omega_0^2}.$$

(b) $KG(s) = \dfrac{K}{(s+10)(s+1)^2}$

(c) $KG(s) = \dfrac{K(s+10)(s+1)}{(s+100)(s+5)^3}$

6.17 Determine the range of K for which each of the listed systems is stable by making a Bode plot for $K = 1$ and imagining the magnitude plot sliding up or down until instability results. Verify your answers by using Matlab with the marginal stability value of K.

(a) $KG(s) = \dfrac{K(s+1)}{s(s+10)}$

(b) $KG(s) = \dfrac{K(s+1)}{s^2(s+10)}$

(c) $KG(s) = \dfrac{K}{(s+2)(s^2+9)}$

(d) $KG(s) = \dfrac{K(s+1)^2}{s^3(s+10)}$

Problems for Section 6.3: The Nyquist Stability Criterion

6.18 **(a)** Sketch the Nyquist plot for an open-loop system with transfer function $1/s^2$; that is, sketch

$$\left.\frac{1}{s^2}\right|_{s=C_1},$$

where C_1 is a contour enclosing the entire RHP, as shown in Fig. 6.17. (*Hint*: Assume C_1 takes a small detour around the poles at $s = 0$, as shown in Fig. 6.27)

(b) Repeat part (a) for an open-loop system whose transfer function is $G(s) = \dfrac{1}{s^2+\omega_0^2}$.

6.19 Sketch the Nyquist plot based on the Bode plots for each of the following systems, then compare your result with that obtained by using the Matlab command nyquist: Don't be concerned with the details of exactly where the curve goes, but do make sure it crosses the real axis at the right spot, has the correct number of -1 encirclements and goes off to infinity in the correct direction.

(a) $KG(s) = \dfrac{K(s+2)}{s+10}$

(b) $KG(s) = \dfrac{K}{(s+10)(s+2)^2}$

(c) $KG(s) = \dfrac{K(s+10)(s+1)}{(s+100)(s+2)^3}$

(d) Using your plots, estimate the range of K for which each system is stable, and qualitatively verify your result by using a rough sketch of a root-locus plot.

6.20 Draw a Nyquist plot for

$$KG(s) = \frac{K(s+1)}{s(s+3)}, \tag{6.73}$$

choosing the contour to be to the right of the singularity on the $j\omega$-axis. Next, using the Nyquist criterion, determine the range of K for which the system is stable. Then redo the Nyquist plot, this time choosing the contour to be to the left of the singularity on the imaginary axis. Again,

using the Nyquist criterion, check the range of K for which the system is stable. Are the answers the same? Should they be?

6.21 Draw the Nyquist plot for the system in Fig. 6.87. Using the Nyquist stability criterion, determine the range of K for which the system is stable. Consider both positive and negative values of K.

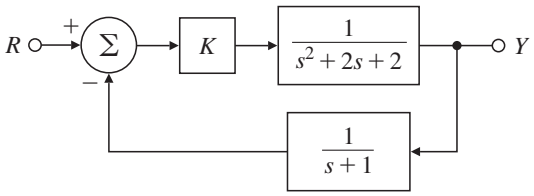


Figure 6.87
Control system for Problem 6.21

6.22 **(a)** For $\omega = 0.1$ to 100 rad/sec, sketch the phase of the minimum-phase system

$$G(s) = \left.\frac{s+1}{s+10}\right|_{s=j\omega}$$

and the nonminimum-phase system

$$G(s) = \left.-\frac{s-1}{s+10}\right|_{s=j\omega},$$

noting that $\angle(j\omega - 1)$ decreases with ω rather than increasing.

(b) Does an RHP zero affect the relationship between the -1 encirclements on a polar plot and the number of unstable closed-loop roots in Eq. (6.28)?

(c) Sketch the phase of the following unstable system for $\omega = 0.1$ to 100 rad/sec:

$$G(s) = \left.\frac{s+1}{s-10}\right|_{s=j\omega}.$$

(d) Check the stability of the systems in (a) and (c) using the Nyquist criterion on $KG(s)$. Determine the range of K for which the closed-loop system is stable, and check your results qualitatively by using a rough root-locus sketch.

6.23 *Nyquist plots and the classical plane curves:* Determine the Nyquist plot, using Matlab, for the systems given below, with $K = 1$, and verify that the beginning point and end point for the $j\omega > 0$ portion have the correct magnitude and phase:

(a) The classical curve called Cayley's Sextic, discovered by Maclaurin in 1718:

$$KG(s) = K\frac{1}{(s+1)^3}.$$

(b) The classical curve called the Cissoid, meaning ivy-shaped:

$$KG(s) = K\frac{1}{s(s+1)}.$$

(c) The classical curve called the Folium of Kepler, studied by Kepler in 1609:

$$KG(s) = K\frac{1}{(s-1)(s+1)^2}.$$

is shown in Fig. 6.89.

(a) Why does the phase start at $-270°$ at the low frequencies?

(b) Sketch the Nyquist plot for $G(s)$.

(c) Is the closed-loop system for the Bode plot shown in Fig. 6.89 stable?

(d) Will the system be stable if the gain is lowered by a factor of 100? Make a rough sketch of a root locus for the system, and qualitatively confirm your answer.

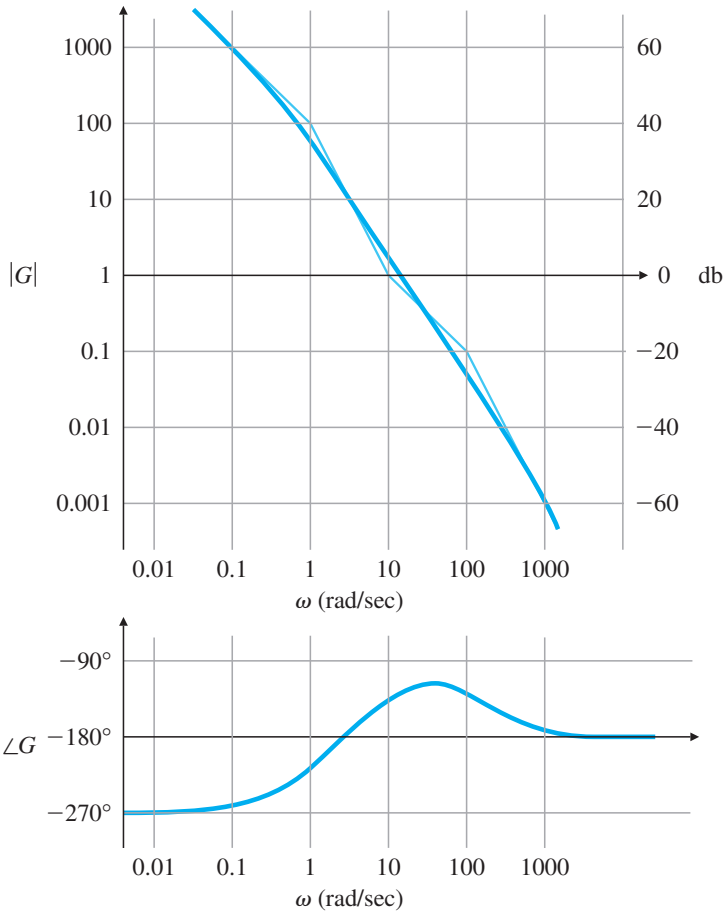


Figure 6.89
Bode plot for Problem 6.25

6.26 Suppose in Fig. 6.90,

$$G(s) = \frac{25(s+1)}{s(s+2)(s^2+2s+16)}.$$

Use Matlab's margin to calculate the PM and GM for $G(s)$ and, on the basis of the Bode plots, conclude which margin would provide more useful information to the control designer for this system.

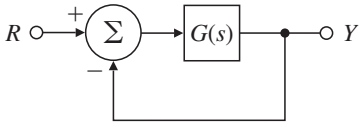


Figure 6.90
Control system for Problem 6.26

6.27 Consider the system given in Fig. 6.91.

(a) Use Matlab to obtain Bode plots for $K = 1$, then use the plots to estimate the range of K for which the system will be stable.

(b) Verify the stable range of K by using margin to determine PM for selected values of K.

(c) Use rlocus to determine the values of K at the stability boundaries.

(d) Sketch the Nyquist plot of the system, and use it to verify the number of unstable roots for the unstable ranges of K.

(e) Using Routh's criterion, determine the ranges of K for closed-loop stability of this system.

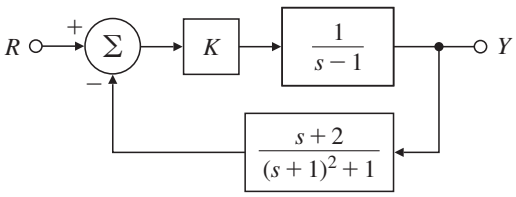


Figure 6.91
Control system for Problem 6.27

6.28 Suppose in Fig. 6.90,

$$G(s) = \frac{3.2(s+1)}{s(s+2)(s^2+0.2s+16)}.$$

Use Matlab's margin to calculate the PM and GM for $G(s)$, and comment on whether you think this system will have well-damped closed-loop roots.

6.29 For a given system, show that the ultimate period P_u and the corresponding ultimate gain K_u for the Ziegler–Nichols method can be found by using the following:

(a) Nyquist diagram

(b) Bode plot

(c) Root locus

6.30 If a system has the open-loop transfer function

$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n)},$$

with unity feedback, then the closed-loop transfer function is given by

$$T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}.$$

Verify the values of the PM shown in Fig. 6.36 for $\zeta = 0.1$, 0.4, and 0.7.

6.31 Consider the unity feedback system with the open-loop transfer function

$$G(s) = \frac{K}{s\left(\frac{s}{0.4}+1\right)\left(\frac{s^2}{4}+\frac{s}{5}+1\right)}$$

(a) Use Matlab to draw the Bode plots for $G(j\omega)$ assuming $K = 1$.

(b) What gain K is required for a PM of 50°? What is the GM for this value of K?

(c) What is K_v when the gain K is set for PM = 50°?

(d) Create a root locus with respect to K, and indicate the roots for a PM of 50°.

6.32 For the system depicted in Fig. 6.92(a), the transfer-function blocks are defined by

$$G(s) = \frac{1}{(s+2)^2(s+4)} \quad \text{and} \quad H(s) = \frac{1}{s+1}.$$

(a) Using rlocus and rlocfind, determine the value of K at the stability boundary.

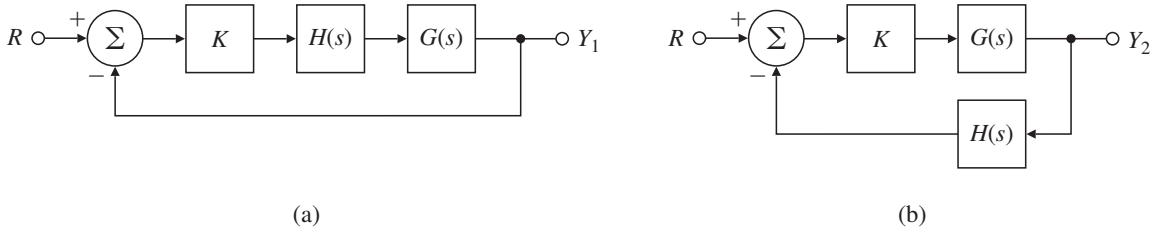
(b) Using rlocus and rlocfind, determine the value of K that will produce roots with damping corresponding to $\zeta = 0.707$.

(c) What is the GM of the system if the gain is set to the value determined in part (b)? Answer this question *without* using any frequency-response methods.

(d) Create the Bode plots for the system, and determine the GM that results for PM = 65°. What damping ratio would you expect for this PM?

(e) Sketch a root locus for the system shown in Fig. 6.92 (b). How does it differ from the one in part (a)?

(f) For the systems in Figs. 6.92 (a) and (b), how does the transfer function $Y_2(s)/R(s)$ differ from $Y_1(s)/R(s)$? Would you expect the step response to $r(t)$ to be different for the two cases?



(a) (b)

Figure 6.92
Block diagram for Problem 6.32: (a) unity feedback; (b) $H(s)$ in feedback

6.33 For the system shown in Fig. 6.93, use Bode and root-locus plots to determine the gain and frequency at which instability occurs. What gain (or gains) gives a PM of 20°? What is the GM when PM = 20°?

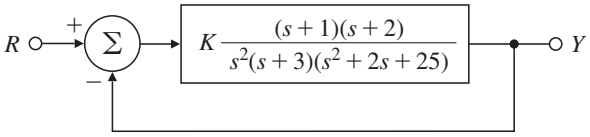


Figure 6.93
Control system for Problem 6.33

Problem for Section 6.6: Closed-Loop Frequency Response

6.43 For the open-loop system

$$KG(s) = \frac{K(s+0.1)}{s^2(s+9)^2},$$

determine the value for K that will yield PM $\geq 30°$ and the maximum possible closed-loop bandwidth. Use Matlab to find the bandwidth.

Problems for Section 6.7: Compensation Design

6.44 For the lead compensator

$$D_c(s) = \frac{T_D s + 1}{\alpha T_D s + 1},$$

where $\alpha < 1$,

(a) Show the phase of the lead compensator is given by

$$\phi = \tan^{-1}(T_D\omega) - \tan^{-1}(\alpha T_D\omega).$$

(b) Show the frequency where the phase is maximum is given by

$$\omega_{\max} = \frac{1}{T_D\sqrt{\alpha}}$$

and the maximum phase corresponds to

$$\sin\phi_{\max} = \frac{1-\alpha}{1+\alpha}.$$

(c) Rewrite your expression for $\omega_{\max}$ to show the maximum-phase frequency occurs at the geometric mean of the two corner frequencies on a logarithmic scale:

$$\log\omega_{\max} = \frac{1}{2}\left(\log\frac{1}{T_D} + \log\frac{1}{\alpha T_D}\right).$$

(d) To derive the same results in terms of the pole–zero locations, rewrite $D_c(s)$ as

$$D_c(s) = \frac{s+z}{s+p},$$

then show that the phase is given by

$$\phi = \tan^{-1}\left(\frac{\omega}{|z|}\right) - \tan^{-1}\left(\frac{\omega}{|p|}\right),$$

such that

$$\omega_{\max} = \sqrt{|z|\,|p|}.$$

Hence, the frequency at which the phase is maximum is the square root of the product of the pole and zero locations.

6.45 For the third-order servo system

$$G(s) = \frac{10{,}000}{s(s+20)(s+30)},$$

design a lead compensator so that $PM \geq 60°$ and $\omega_{BW} \geq 10$ rad/sec using Bode plot sketches.

6.46 For the system shown in Fig. 6.100, suppose

$$G(s) = \frac{5}{s(s+1)(s/5+1)}.$$

Use Bode plot sketches to design a lead compensation $D_c(s)$ with unity DC gain so that PM $\geq 40°$. Then verify and refine your design by using Matlab. What is the approximate bandwidth of the system?

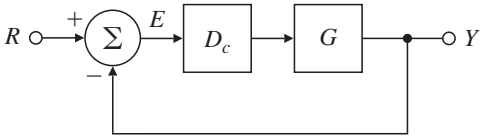


Figure 6.100
Control system for Problem 6.46

6.47 Derive the transfer function from T_d to θ for the system in Fig. 6.67. Then apply the Final Value Theorem (assuming T_d = constant) to determine whether $\theta(\infty)$ is nonzero for the following two cases:

(a) When $D_c(s)$ has no integral term: $\lim_{s\to 0} D_c(s) =$ constant;

(b) When $D_c(s)$ has an integral term:

$$D_c(s) = \frac{D_c'(s)}{s}$$

In this case, $\lim_{s\to 0} D_c'(s) =$ constant.

6.48 The inverted pendulum has a transfer function given by Eq. (2.31), which is similar to

$$G(s) = \frac{1}{s^2 - 1}.$$

(a) Use Bode plot sketches to design a lead compensator to achieve a PM of 30°. Then verify and refine your design by using Matlab.

(b) Sketch a root locus and correlate it with the Bode plot of the system.

(c) Could you obtain the frequency response of this system experimentally?

6.49 The open-loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s\left(\frac{s}{3}+1\right)\left(\frac{s}{40}+1\right)}.$$

(a) Design a lag compensator for $G(s)$ using Bode plot sketches so that the closed-loop system satisfies the following specifications:

(i) The steady-state error to a unit ramp reference input is less than 0.05.

(ii) PM $\geq 40°$.

(b) Verify and refine your design by using Matlab.

6.50 The open-loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{(s+10)\left(\frac{s^2}{5} + \frac{s}{6} + 1\right)}.$$

(a) Design a lead compensator for $G(s)$ using Bode plot sketches so that the closed-loop system satisfies the following specifications:

(i) The steady-state error to a unit step reference input is less than 0.01.

(ii) For the dominant closed-loop poles the damping ratio $\zeta \geq 0.4$.

(b) Verify your design with a direct computation of the damping of the dominant closed-loop poles.

6.51 A DC motor with negligible armature inductance is to be used in a position control system. Its open-loop transfer function is given by

$$G(s) = \frac{37}{s\left(\frac{s}{7} + 1\right)}.$$

(a) Design a compensator for the motor using Bode plot sketches so that the closed-loop system satisfies the following specifications:

(i) The steady-state error to a unit ramp input is less than $1/150$.

(ii) The unit step response has an overshoot of less than 20%.

(iii) The bandwidth of the compensated system is no less than that of the uncompensated system.

(b) Verify and/or refine your design including a direct computation step response overshoot.

6.52 The open-loop transfer function of a unity feedback system is

$$G(s) = \frac{K}{s\left(1 + \frac{s}{0.5}\right)\left(1 + \frac{s}{1.5}\right)}.$$

(a) Sketch the system block diagram including input reference commands and sensor noise.

(b) Design a compensator for $G(s)$ using Bode plot sketches so that the closed-loop system satisfies the following specifications:

(i) The steady-state error to a unit ramp input is less than 0.03.

(ii) PM $\geq 45°$.

(iii) The steady-state error for sinusoidal inputs with $\omega < 0.02$ rad/sec is less than $1/300$.

(iv) Noise components introduced with the sensor signal at frequencies greater than 50 rad/sec are to be attenuated at the output by at least a factor of 1000.

(c) Verify and/or refine your design including a computation of the closed-loop frequency response to verify (iv).

6.53 The transfer function for a quadrotor attitude control system between a pitch control input, T_{lon}, and the pitch angle, θ, is

$$\frac{\theta(s)}{T_{lon}(s)} = G_1(s) = \frac{1}{s^2(s+2)}.$$